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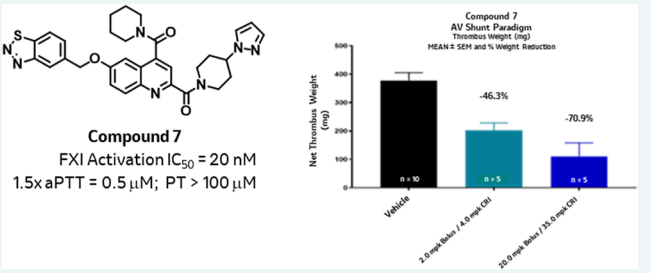
The Discovery of a Novel Anticoagulant Mechanism: Factor XI Activation Inhibitors

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ABSTRACT: Despite advances in anticoagulant drug discovery to establish a new standard of care, represented by Factor Xa (FXa) inhibitors, additional therapies are needed to address the concomitant bleeding risk posed by these agents. Factor XI (FXI) plays a key role in the intrinsic coagulation cascade contributing to thrombin generation and fibril formation, and there is an increasing weight of evidence supporting a comparable efficacy and superior safety profile of FXI-based oral anticoagulants. A novel allosteric mechanism of action (MoA) for inhibiting FXI activation to Factor XIa (FXIa) was discovered by applying an AS-MS-based screen to the FXI zymogen. Biochemical and biophysical characterization of these structurally diverse small-molecule classes confirmed this mode of inhibition as specific to the FXI zymogen, and a high-resolution FXI-inhibitor cocomplex structure identified the binding pocket that is distal to the FXI cleavage site. These FXI Activation Inhibitors were active in translational functional assays, prolonging activated partial thromboplastin time (aPTT) in human plasma and demonstrating dose-dependent antithrombotic effects in the rabbit arteriovenous-shunt model of thrombosis. The results affirm that FXI Activation Inhibitors are a novel mechanism for the potential treatment of thrombotic disorders.



KEYWORDS: *factor XI, coagulation, thrombosis*

INTRODUCTION

Cardiovascular disease represents a leading cause of mortality worldwide.[1−4] Thrombosis[5] is a key contributor of the underlying pathology associated with ischemic heart disease,[6] stroke,[7] and venous thromboembolism (VTE)[8] and, consequently, presents a significant risk factor for cardiovascular-related deaths. The evolution of oral anticoagulant therapies from warfarin and nonvitamin K antagonist to thrombin and Factor Xa (FXa) inhibitors has progressively improved the therapeutic index.[9−11] However, despite these advances, a significant bleeding risk liability persists, thereby necessitating a need for alternative anticoagulant therapies that further minimize their impact on hemostasis.[12,13]

Factor XI is a circulating pre-enzyme within the intrinsic coagulation cascade that is cleaved by Factor XIIa and undergoes a conformational change to generate catalytically active Factor XIa, thereby propagating the aforementioned intrinsic coagulation cascade, ultimately leading to thrombin activation.[14−16] In addition to a catalytic domain, the Factor XI/XIa protein is also comprised of four distinct regulatory Apple domains that function independently to modulate the binding of Factor XI/XIa to cofactor high-molecular-weight kininogen (HMWK) by Apple 2, effector Factor XIIa by Apple 4, substrate Factor IX by Apple 3, and feedback effector thrombin by Apple 1, respectively (Figure 1).[17−19] In addition to the orthosteric inhibition of the Factor XIa catalytic domain, suppressing the necessary trafficking functions of any Apple domain could provide an additional pharmacologically relevant inhibitory mechanism.

Significant preclinical and clinical evidence supports Factor XI (FXI) as an efficacious and safe alternate target within the intrinsic coagulation cascade[16] (Figure 1). Human epidemiological data sets suggest that patients with congenital FXI deficiency (hemophilia C) present a low bleeding risk,[20] a finding that is also corroborated pharmacologically in rodent in vivo models.[21] The breadth of preclinical and clinical evidence has triggered a significant effort in the discovery of novel FXI-

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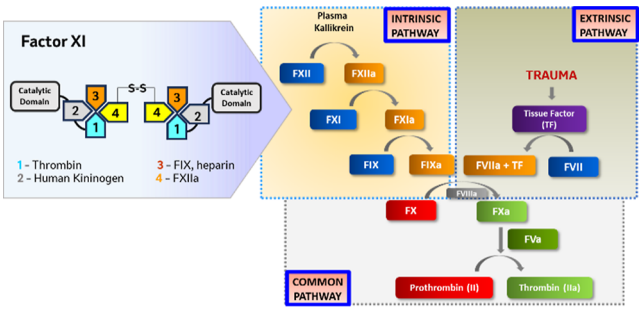


Figure 1. Illustration of the intrinsic and extrinsic coagulation cascades. Shown at left is a cartoon figure that illustrates the FXI zymogen homodimer.

directed therapies.[22] Early advanced FXI clinical data was reported for an antisense oligonucleotide (ASO), BAY-2306001,[23] wherein FXI knockdown subsequently reduced the risk of postoperative VTE in patients undergoing elective arthroplasty. Additional Phase II clinical study of BAY-2306001 in patients with end-stage renal disease demonstrated that a dose-dependent reduction in circulating FXI levels led to a corresponding reduction in clotting events.[24] Although development of BAY-2306001 was halted in 2021, a different ASO, fesomersen, has progressed into Phase 2, as have several FXI-specific monoclonal antibodies (mAbs), osocimab,[25] MK-2060, and xisomab 3G3.[26] Further, osocimab has entered Phase 3, having demonstrated noninferiority to standard of care for the treatment of VTE.[25]

Despite the acceptable safety and tolerability of FXI-directed ASO and mAbs, the clinical utility of these nonsmall-molecule modalities, with their slow onset of action and/or long duration of action, is yet to be determined. In contrast, small-molecule FXIa inhibitors provide flexibility with oral delivery option, rapid onset, short half-life along with the potential to be used in both acute and chronic settings.[21,27] As such, significant research in the biopharmaceutical sector has sought to advance FXIa small-molecule inhibitors with asundexian and milvexian having progressed into Phase 3 clinical development.[28,29] The most advanced of these FXIa inhibitors, asundexian, had entered a Phase 3 study for stroke prevention; however, the trial was terminated early due to inferior efficacy.[28,30,31] A different FXIa inhibitor, milvexian, was shown to reduce venous and arterial thromboembolism in a similar knee replacement surgery group as described above and has entered a Phase 3 outcome trial seeking to reduce a composite end point of stroke and systemic embolism with reduced bleeding.[29,32,33] Numerous additional publications report on the discovery and characterization of orthosteric and allosteric inhibitors of FXIa that, despite their elegance, illustrate the challenges associated with optimizing these molecules for their requisite affinity, selectivity, and/or oral bioavailability.[34−38]

We hypothesized that small-molecule inhibitors of FXI activation operating in an FXI-dependent and zymogen-specific manner would represent a pharmacologically differentiated MoA from the classical orthosteric FXIa inhibition mechanism of action. Such compounds would exhibit function and specificity traditionally reserved for biologics like mAbs while retaining the desirable small-molecule drug-like properties. To the best of our knowledge, no such FXI Activation Inhibitors (XIAIs) have been described in the literature.

AS-MS technology[39,40] represents a powerful platform to screen for novel mechanisms by enabling screening of large mixture-based combinatorial libraries (5.5 M compound collection) in a label-free high-throughput format that has proven to be highly reproducible. Screening against the FXI zymogen instead of an activated full length FXIa protein represents a subtle, but critical, way to identify novel mechanisms for inhibiting FXI pharmacology, as it excludes conformational artifacts associated with catalytic domain activation. Execution of the primary screen (Figures 2 and

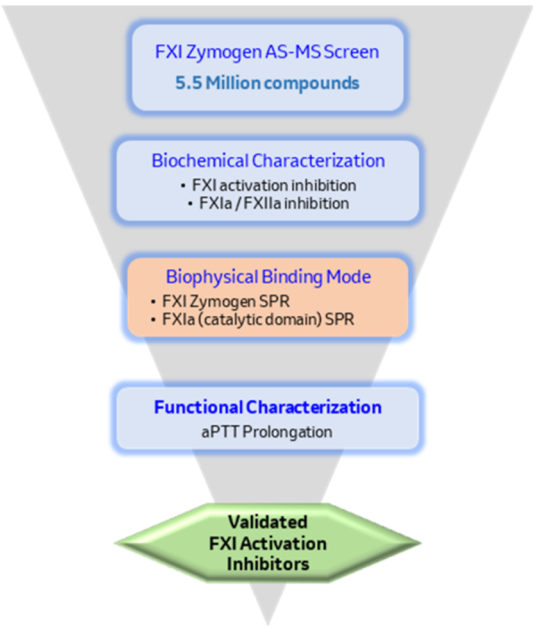


Figure 2. FXI zymogen screening plan.

S1) involved incubation of the FXI zymogen with mass-encoded combinatorial libraries followed by automated microscale size-exclusion chromatographic separation of the small molecules from the protein. This was followed by reverse-phase LCMS and deconvolution of each mass-encoded mixture to identify hits from the primary screen.[39,40]

Translation of binding to functional inhibition of FXI Activation required the development of a coupled enzymatic assay, wherein FXIIa-catalyzed FXI activation was measured via FXIa substrate turnover. Preincubation of binding hits with the FXI zymogen was followed by the addition of catalytically active FXIIa to evaluate the functional activity for each hit. The addition of a FXIIa-specific inhibitor as a stop reagent, followed by a fluorescent peptide FXIa substrate[41] enabled measurement of the desired pharmacology.

MATERIALS AND METHODS

General Methods—Chemical Synthesis. All reagents were commercially available and used without further purification, unless otherwise stated. Column chromatography was carried out on flash silica gel (Merck KGaA 230−400 mesh). TLC analysis was conducted on Analtech silica gel plates. The LCMS analyses were performed using a Micromass ZMD mass spectrometer coupled to an Agilent 1100 Series HPLC utilizing a YMC ODS-A 4.6 mm × 50 mm column eluting at 4.5 mL/min with a solvent gradient of 10 to 95% B over 2.5 min, followed by 0.5 min at 95% B: solvent A = 0.06% TFA in water; solvent B = 0.05% TFA in acetonitrile. ^{1}H NMR

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spectra were obtained on 500 or 600 MHz Varian or Bruker Spectrometers in $CDCl_3$, CD_3OD, or DMSO-d_6 as indicated, chemical shifts are reported as δ using the solvent peak as the reference, and coupling constants are reported in hertz (Hz). All the assayed compounds described in this manuscript were at least 95% pure as judged by LCMS and NMR.

Compounds 1−4. Compounds **1** through **4** were repurified from either solid samples or 10 mM DMSO screening solutions using preparative Reverse-Phase HPLC (acetonitrile/H_2O + 0.1% TFA additive). Representative spectra confirming their structure and purity are presented in the Supporting Information.

Synthesis of Compounds 5−7. Compounds **5**−**7** were prepared following similar procedures outlined in a representative synthesis for Compound **5** within the Supporting Information. Representative spectra confirming their structure and purity are presented therein.

General Methods—In Vitro Assays. Full-length human Factor XI zymogen was purchased from Haematologic Technologies Inc. (catalog number HCXI-150). Full-length Factor XIa was purchased from Sekisui Diagnostics (catalog number 4011A). Human full-length Factor XIIa and human HMWK were purchased from Enzyme Research Laboratories (catalog numbers HFXIIa and HK, respectively). Substrate z-GPR-AFC was purchased from Sigma (catalog no. C0980-10MG), and substrate ch-GGR-AFC was purchased from CPC Scientific (catalog number 839493).

For human subjects, samples were obtained through the informed consent of donors (WCG IRB; Protocol ID: 1133978; Approval Date: 07/09/2024). Please note the Protocol is not specific to the research related to this publication but rather a general protocol for the human sample collection.

In Vitro Assays (FXIa, Plasma Kallikrein, Factor XIIa, Factor IXa). Compounds were preincubated for 30 min at 25 °C with a human enzyme (0.04 nM Factor XIa, 0.05 nM kallikrein, 1 nM Factor XIIa, or 2 nM Factor IXa) in 50 mM HEPES buffer with 150 mM sodium chloride, 5 mM calcium chloride, 0.1% PEG 8000, pH 7.4. Enzymatic activity was determined by addition of a tripeptide substrate and measurement of the fluorescence at 400/505 nm after a 60 min incubation at 25 °C. The Factor XIa assay utilized glycine-proline-arginine-7-amido-4-trifluoromethylcoumarin (GPR-AFC). The Plasma Kallikrein and Factor XIIa assays utilized lysine-proline-arginine-7-amido-4-trifluoromethylcoumarin (KPR-AFC), and the Factor IXa assay utilized cyclohexylglycine-glycine-arginine-7-amido-4-trifluoromethylcoumarin (CHG-Gly-Arg-AFC). The percent inhibition for each data point was calculated from the data and analyzed using the log(inhibitor) vs response four parameters equation to determine the half-maximal inhibitory concentrations (IC_{50}). The IC_{50} values were converted to equilibrium inhibitory constants (K_i) using the Cheng−Prusoff equation.

Activation of Human Factor XI to XIa by Human Factor α-XIIa. The assay was performed in 50 mM HEPES, 150 mM NaCl, 5 mM $CaCl_2$, 0.1% PEG-8000, pH 7.4 buffer in black, flat-bottom, polystyrene microplates (Corning, Cat #3573). Serial dilutions (10-point, 3.333-fold) of the test compounds delivered as 10 mM DMSO stocks were prepared in the Labcyte Echo-qualified PP 384 well microplates (Labcyte, Cat #P-05525). Human coagulation factor (F)XI (Haematologic Technologies Inc., Cat #HCXI-150, concentration 46 nM, 19.5 mL) was preincubated with 0.5 mL of the test compounds for 30 min at 22 °C. The activation was initiated by addition of 10 mL of human coagulation factor (F)XIIa (Enzyme Research Laboratories, Cat #HFXIIa, concentration 30 nM) and reaction proceeded for 60 min at 22 °C, after which it was quenched by addition of a selective inhibitor of FXIIa (Compound 54 in International Patent Publication WO2018093695, concentration 600 mM in DMSO, 1 mL). Ten min after the quench, 29 mL of the z-GPR-AFC substrate (Sigma, Cat #C0980-10MG, concentration 200 μM) was added into each well and the detection reaction proceeded for 60 min at 22 °C. Fluorescence at 405/510 nm was measured in end-point mode using an Envision plate reader (PerkinElmer). IDBS Activity Base XE (ABase) analysis using 4 parameter logistic fit was performed to determine Minimum, Maximum, EC_{50}, and Slope Factor for each compound. Relevant IC_{50} data for the FXIIa inhibitor: FXIIa IC_{50} = 58 nM; FXIa IC_{50} = >5000 nM; FIXa IC_{50} = >15 mM.

Activation of Human Factor XI to XIa by Human Factor α-XIIa, HMW Kininogen, and Ellagic Acid. The assay was performed in 50 mM HEPES, 150 mM NaCl, 5 mM $CaCl_2$, 0.1% PEG-8000, pH 7.4 buffer in black, flat-bottom, polystyrene microplates (Corning, Cat #3573). Serial dilutions (10-point, 3.333-fold) of the test compounds delivered as 10 mM DMSO stocks were prepared in the Labcyte Echo-qualified PP 384 well microplates (Labcyte, Cat #P-05525). 19.5 mL of human coagulation factor (F)XI (Haematologic Technologies Inc., Cat #HCXI-150, concentration 46 nM) and single-chain HMWK (Enzyme Research Laboratories, Cat #HK 1300) were preincubated with 0.5 mL of the test compounds for 30 min at 22 °C. The activation was initiated by addition of 10 mL of activation mixture containing the human coagulation factor (F)XIIa (Enzyme Research Laboratories, Cat #HFXIIa, concentration 0.15 nM) and APTT-XL Pacific Hemostasis Activated Partial Thromboplastin Time (aPTT) Reagent containing ellagic acid (Thermo Scientific, Cat #100403, diluted 16.67-fold). Reaction proceeded for 60 min at 22 °C, after which it was quenched by addition of a selective inhibitor of FXIIa (Compound 54 in International Patent Publication WO2018093695, concentration 600 mM in DMSO, 1 mL). Ten min after the quench, 29 mL of the z-GPR-AFC substrate (Sigma, cat. no. C0980-10MG, concentration 200 mM) was added into each well, and the detection reaction proceeded for 60 min at 22 °C. Fluorescence at 405/510 nm was measured in end-point mode using an Envision plate reader (PerkinElmer). IDBS Activity Base XE (ABase) analysis using 4 parameter logistic fit was performed to determine Minimum, Maximum, EC_{50}, and Slope Factor for each compound. Relevant IC_{50} data for the FXIIa inhibitor: FXIIa IC_{50} = 58 nM; FXIa IC_{50} = >5000 nM; FIXa IC_{50} = >15 mM.

Activated Partial Thromboplastin Time (aPTT) and Prothrombin (PT) Assay. Activated partial thromboplastin time (aPTT) is a clotting test that measures the intrinsic coagulation cascade. The test is performed in sodium citrated plasma. Rabbit plasma was purchased from Lampire Biologics and human plasma was made by collecting blood from healthy donors of both genders into Na citrate tubes (Sarstedt coagulation 9NC/10 mL). Blood was centrifuged at 1500*g* and the plasma collected. aPTT values were checked on each individual donor, and those within the normal range (28−40 s) are pooled, aliquoted, and stored at −80 °C. Test samples were prepared by spiking inhibitors or vehicle into plasma. These

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spiked samples were then run on a coagulation analyzer (STA-R Evolution, Stago Diagnostica). In general, the analyzer performs the following steps: Factor XII is activated by addition of ellagic acid (Pacific Hemostasis), and then time to clot is measured after recalcification of the sample. Inhibition of FXI will cause the aPTT clot time to be prolonged. The data are expressed as percent increase over vehicle control clot time, and the concentration that causes a 50% increase (1.5×) of clot time is reported. The PT assay is conducted in a similar fashion, substituting the tissue factor (Pacific Hemostasis) for ellagic acid to activate Factor VII.

Expression and Purification of the FXIa-Catalytic Domain for SPR Experiments. The DNA encoding the residues for the FXIa catalytic domain (388−625) with single mutation C500S was cloned into the pGLY4265 for transformation into *Pichia pastoris*, host X-33 strain. After transformation, cells were cultured on YPDS-agar plates with Zeocin selection for 3−4 days. Clones were selected for small-scale expression in BMMY culture, and expression was evaluated by SDS-PAGE and Western blot. The best-expressing clone was then scaled up to 20 L of expression for purification. Preinduced cultures are grown in BMGY media for 72 h. The preinduced cultures are then resuspended in BMMY media with 1% methanol (inducer) at 2× the volume of the initial culture. The cells are expressed for 40 h post induction, with methanol added to 1% every morning and afternoon to avoid evaporation. After expression, the secreted media are concentrated by TFF and then purified by Ion Exchange Chromatography in the presence of 1 mM Benzamidine. The pooled protein is then treated with EndoHf overnight at 8 °C. The extent of deglycosylation was determined by SDS-PAGE. The protein was then captured using a Heparin sepharose column followed by final polishing with a gel filtration step (Superdex75 16/60). The protein was concentrated to 23.9 mg/mL in an approximate yield of 3.2 mg/L.

Expression and Purification of the FXI-Apple Protein (No Catalytic Domain) for SPR Experiments. The DNA encoding amino acid sequence of the monomeric Apple domain variant FXIa(1-378; C339S, Y347A, K349A)-TEV-8His (with no catalytic domain present) was cloned into the pcDNA 3.1_vector. The plasmid was transfected into Expi-CHO and cultured for 72 h. The secreted protein was then harvested by centrifugation and concentrated by TFF. The concentrated medium was clarified and purified by IMAC followed by overnight TEV protease cleavage of the C-terminal His tag. The extent of cleavage was determined by SDS-PAGE and Western blot. Following the cleavage, the protein was further purified by capture with a Heparin column. To ensure that there was no carry over of any uncleaved material, we performed a subtractive IMAC followed by a final gel filtration step (Superdex200 16/60). Overall expression yield was 0.36 mg/L and the final protein was delivered at 8.99 mg/mL with estimated >95% purity.

Titration of Compounds in BIAcore SPR with Immobilized Tool Proteins. BIAcore CM5 sensor chip (GE, cat. no. BR-1005-30) was docked into the BIAcore T100 (with T200 software) system (GE) and equilibrated with 3 repeats of the priming step with 1× HBS-N, 0.005% P-20 immobilization buffer. Immobilization of ∼3,00 RU's of FXI, FXIa, FXIIa, Prekallikrein (commercially sourced), FXIa-Catalytic domain, and FXI-Apple domains (prepared internally) was carried out using a standard Biacore procedure employing amine coupling chemistry followed by the deactivation with ethanolamine. 1× BIAcore running buffer (1× HBS-N, 5 mM $CaCl_2$, 0.005% P-20, 2% DMSO) was prepared for testing of binding of compounds of interest. Compounds, delivered as 10 mM DMSO stocks, were first diluted 20-fold with DMSO to 500 mM concentration and then 50-fold (10 mM final compound concentrations) into 1.02× BIAcore running buffer lacking DMSO to correct the final DMSO concentration to 2%. 5-Point, 3-fold serial dilutions with 1× BIAcore running buffer were prepared. Samples were injected at 30 mL/min on the chip with 60 s association and 180 s dissociation time. Solvent correction was performed before sample injections began, after every 192 cycles, and at the end of the experiment. Data analysis was carried out using BIAevaluation software, with Ligand Injections corrected using referencing to flow cell 1 and fitted with a 1:1 binding model to determine binding constants.

Expression and Purification of the FXI Zymogen Protein for X-ray Crystallography. The DNA encoding the full-length amino acid sequence of the FXI(1−625) was cloned into the pcDNA 3.4_vector. The plasmid was transfected into Expi-CHO and cultured for 72 h. The secreted protein was then harvested by centrifugation and concentrated by TFF. The concentrated medium was clarified and captured using a Heparin HP column in the presence of 1 mM Benzamidine. NHSO4 was added to the pooled protein to 1 M. The pool was cloudy after $(NH_4)_2SO_4$ addition and clarified prior to loading onto a Butyl HP column for Hydrophobic interaction Chromatography followed by a final polishing step with a gel filtration step (Superdex 200 16/60). The protein was concentrated to 9.22 mg/mL with an approximate yield of 0.38 mg/L with an estimated 90−95% purity.

X-ray Crystallography Methods. FXI Zymogen in complex with the HMWK peptide (FNPISDFPDTTS; Peptek; product number 778901558-1) was prepared as 100 mM stock solution of the HMWK peptide was added to FXI at a 10:1 molar ratio and 1 mM of compound 7 from a 100 mM DMSO stock solution. One microliter portion of protein/peptide/ligand complex was added to 1 mL of reservoir solution and incubated at 18 °C against 100 mM HEPES, pH 7.3, 17−20% PEG 1500. A single crystal mounted and frozen in liquid nitrogen using 25% glycerol supplement to the reservoir solution was used to collect the data at the Canadian Light Source using cryogenic temperature. The images were processed by AutoProc and an initial molecular replacement solution was found using PDB 5I25 as a starting model by MOLREP. Further refinement was performed using Autobuster, followed by rebuilding in COOT.

General Methods—In Vivo Models. The care of animals and experimental protocols were reviewed and approved by the Merck & Co., Inc., Rahway, NJ 07065, USA, Institutional Animal Care and Use Committee.

AV Shunt: Evaluate the Effect of the Test Compound on the Thrombus Weight and Coagulation Biomarker aPTT and Plasma PK in Anesthetized NVW Rabbits. *Animal Preparation.* Rabbits (NZW, adult) were sedated with ketamine/xylazine cocktail, weighed, and anesthetically maintained with ketamine/xylazine cocktail for the duration of the study period. A dedicated peripheral intravenous catheter was placed percutaneously for fluid therapy and dose administration. Animals then underwent routine surgical isolation and catheterization of the left femoral artery and right femoral vein in preparation for periodic extracorporeal shunting of

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blood and arterial blood collection as per the study timeline. Following femoral catheter instrumentation, animals were equilibrated for 20 min prior to further study manipulations.

Dosing and Extracorporeal Shunt Procedures. Following animal equilibration and baseline arterial blood collection, animals were first dosed with vehicle control (35% HPβCD) delivered via IV bolus over 5 min@2.0 mL/kg, followed by IV constant rate infusion dosing at a rate of 2.0 mL/kg/h via the indwelling peripheral intravenous catheter over 55 min. An AV shunt was connected to the left femoral artery and right femoral vein 20 min post dose initiation permitting 40 min of extracorporeal blood circulation prior to shunt disconnect for thrombus assessment and IV dose infusion discontinuation. Upon disconnection, the shunt was dismantled for thrombus collection and quantification using an analytical scale. Animals were then dosed with the compound in 35% HPβCD via IV bolus over 5 min@2.0 mL/kg, followed by IV constant rate infusion dosing at a rate of 2.0 mL/kg/h via the indwelling peripheral intravenous catheter over 55 min. A new AV shunt was again connected to the left femoral artery and right femoral vein 20 min post dose initiation permitting 40 min of extracorporeal blood circulation prior to shunt disconnection for thrombus assessment and IV dose infusion discontinuation. Upon disconnection, the shunt will be dismantled for thrombus collection and quantification.

RESULTS

The AS-MS hits (annotated henceforth as FXI Activation Inhibitors or XIAI) were found to inhibit FXIIa-mediated conversion of FXI to FXIa, with sub-μM $IC_{50FXI \rightarrow FXIa}$ (Figure 3), with no significant independent effect on either FXIa or

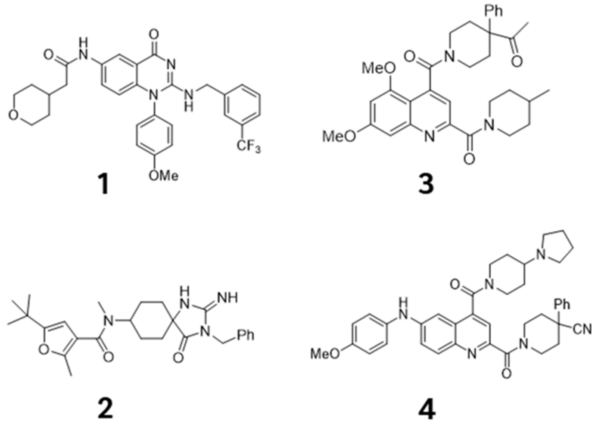


Figure 3. Screening hits from the AS-MS screen that inhibited FXI activation: Structures of screening hits **1**−**4** exhibited functional inhibition of FXI activation by FXIIa.

FXIIa catalytic activity (IC_{50FXIa} and $IC_{50FXIIa}$ >15 μM, Table 1). This screening campaign resulted in the identification of four primary hit compounds from three structurally distinct classes as Factor XI Activation Inhibitors (XIAIs) that likewise exhibited excellent selectivity against other proteases associated with the intrinsic or extrinsic coagulation cascades (Table S1).

Biochemical characterization of each XIAI compound demonstrated functional effects solely directed against FXI zymogen activation, with no observed inhibition of FXIa catalytic activity. To confirm the zymogen-binding specificity, FXI XIAIs were studied using surface plasmon resonance (SPR) yielding equilibrium dissociation constants (K_D) for

Table 1. Dose Response Inhibition Data for Compounds 1−4 Associated with FXI Activation Inhibition and Direct Orthosteric FXIa and FXIIa Inhibition, Respectively

biochemical assay	FXI IC_{50} (μM)	FXIa IC_{50} (μM)	FXIIa IC_{50} (μM)
1	0.3	>15	>15
2	0.04	>15	>15
3	0.07	>15	>15
4	0.06	>15	>15

XIAI binding in the sub-μM range (0.1−0.4 μM) and of comparable potency (Figure 4 and Table 2). However, there were notable differences in the FXI-XIAI kinetic profile for each respective XIAI hit compound, as compounds **1** and **2** bound the FXI with a fast on/fast off kinetics with k_{off} = 0.15 and 0.04 s^{-1}, while compounds **3** and **4** demonstrated slow on/slow off kinetics with k_{off} = 0.008 and 0.007 s^{-1}, respectively. SPR characterization of the four XIAI compounds with full-length FXIa indicated comparable saturable binding with the FXI zymogen. However, similar SPR-binding experiments between the XIAIs and the truncated FXIa catalytic domain alone resulted in no observed protein binding (Table 2). In addition, a counterscreen of the XIAIs with FXIIa and plasma kallikrein, a closely related homologue of FXI, showed no evidence of specific binding under similar conditions. Taken together, these data sets support a mechanism whereby the XIAIs exert their effect by binding within the regulatory subunit, and unlike other reported allosteric FXIa inhibitors, do not engage the active site within the FXIa catalytic domain.

Almost all FXI zymogen that is circulating in the plasma is complexed with HMWK, a highly glycosylated plasma protein.[18] X-ray studies have confirmed binding of HMWK to the FXI Apple 2 domain within the regulatory subunit. To understand if XIAIs can engage the FXI−HMWK complex and retain inhibitory effects on FXI to FXIa conversion, we measured $IC_{50FXI \rightarrow FXIa}$ in the presence of saturating amounts of the HMWK protein. Compounds **1**−**4** showed dose-dependent inhibition of FXIIa-mediated conversion of FXI to FXIa in the presence of HMWK. The data confirmed that precomplexation of FXI−HMWK does not impact the inhibition profiles of these compounds and further rules out any effect XIAIs may have in modulating FXI−HMWK dynamics as a potential mechanism of action.

To assess functional effects of XIAIs, a more physiologically relevant context, we studied anticoagulant activity in human plasma using prothrombin and activated partial thromboplastin time assays (PT and aPTT, respectively).[42,43] A dose-dependent prolongation of aPTT was observed for compounds **3** and **4**, while no significant effect on aPTT was observed for compounds **1** and **2** (Table 3). The 150% (or 1.5×) increase in aPTT was not measurable under the conditions tested; however, 22% and 37% increases in aPTT were observed for compounds **3** and **4** at 100 μM concentrations, respectively. Corroborating the FXI-specific pharmacology exhibited by the XIAIs, no significant impact on the PT was observed under similar conditions.

Hit expansion of compounds **1**−**4** based on 2D-computational structural similarity yielded structure−activity relationships (SARs) that suggested that each hit was progressible based on FXI Activation inhibition, but that only quinoline diamides **3** and **4** yielded SAR that additionally resulted in prolongation of clotting time in the aPTT assay.

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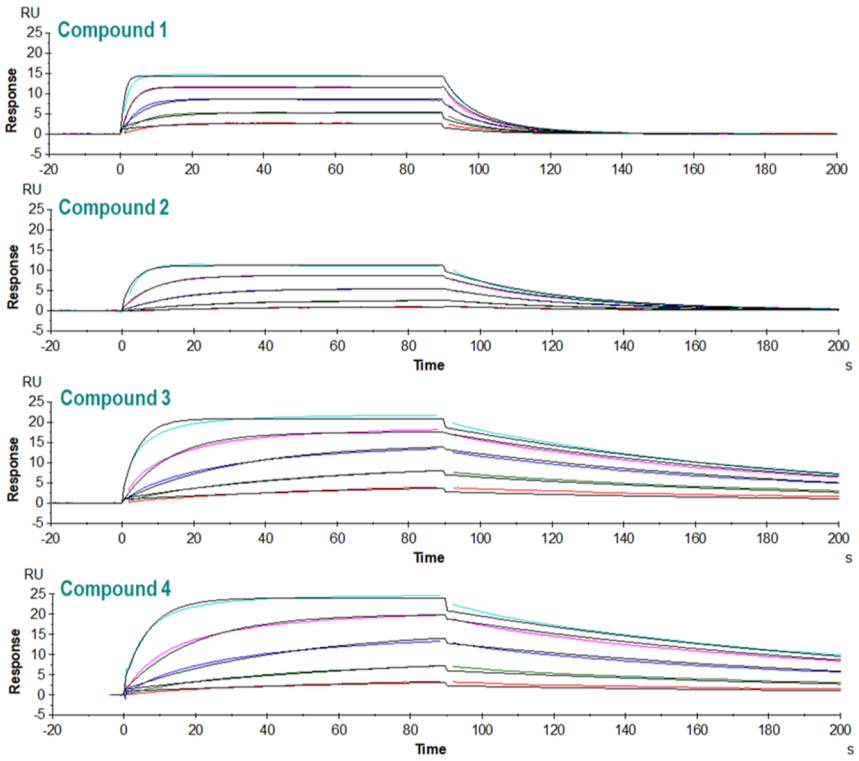


Figure 4. SPR evaluation of XIAI hits with immobilized full-length FXI to determine if each exhibited reversible and/or saturable binding.

Table 2. In Vitro Binding Constants for XIAI Hits to Full-Length FXI and FXIa, the FXIa Catalytic Domain (Only)

	FXI		FXIa		FXIa catalytic domain	
compound	K_d (nM)	k_{off} (1/s)	K_d (nM)	k_{off} (1/s)	K_d (nM)	k_{off} (1/s)
1	132	0.15	98	0.06	>10,000	N/A
2	134	0.04	145	0.02	>10,000	N/A
3	408	0.008	512	0.008	>10,000	N/A
4	508	0.007	933	0.008	>10,000	N/A

Table 3. aPTT and PT Evaluation of XIAI Hits[a]

	aPTT		PT	
compound	% change @100 mM (%)	1.5× conc. (μM)	% change @100 mM (%)	1.5× conc. (μM)
1	7	>100	−1	>100
2	7	>100	0	>100
3	22	>100	0	>100
4	37	>100	2	>100

[a]Characterization of the XIAI hit compounds for functional inhibition of coagulation based on initiation of the intrinsic (aPTT) and extrinsic (PT) coagulation pathways. Standard deviation for aPTT measurements (N = 3): **1**, ±2.0%; **2**, ±0.8%; **3**, ±2.0%; **4**, ±1.9%.

Basic hit-to-lead efforts were initiated to confirm common convergent SAR for each hit and to subsequently ascertain the tractability for each series. Early SAR, derived from both screening existing similar analogs from our internal sample collection, as well as a limited forward medicinal chemistry effort, affirmed that **3** and **4** both did indeed converge on a similar 6-substituted quinoline diamide motif.[44]

A preliminary dive into structural requirements indicated that the C2- and C4-amides are tertiary with a strong preference for cyclic structures. In addition, while it appeared that substitution on the C2-amide ring system could access greater potency in the FXI Activation Assay, there was an opposing preference for unsubstituted cyclic amides at C4.

Variation of the C6-substituent was determined to have a significant effect on driving potency, in addition to favorably modulating aPTT activity, as replacement of the aryl amine substituent from **4** with a benzyloxy moiety (compound **5**) led to a profound improvement in the more translationally relevant 1.5× aPTT activity. Continued Hit-to-Lead SAR identified structural changes that could differentially impact aPTT potency without a parallel improvement in FXI Activation potency (compound **6**).

Early medicinal chemistry efforts directed toward identifying a minimal pharmacophore initially focused on simplification of the C2-amide heterocycle distal quaternary center, which resulted in the discovery of compound **7** that retained previous advances in aPTT activity while driving an improvement in physical properties[46] (Tables 4 and S2). In addition to retaining aPTT activity, compound **7** exhibited pharmacokinetic properties suitable for profiling in an acute PD efficacy model that would provide in vivo validation for the XIAI mechanism (Table S2).

Concurrent with continued biophysical characterization of the nascent XIAIs, efforts to identify the putative binding pocket and, hence, inform on mechanistic hypotheses continued. More detailed granularity and confirmation of mechanism would likely require direct characterization of an FXI-bound XIAI molecule. During the course of SAR development, the cocrystal structure of a simplified pyridyl diamide core compound **7** was solved and revealed the binding site to be within the regulatory domain, as previously suggested (Figure 5). An intriguing conclusion was the apparent induced fit of compound **7** at the junction of all four Apple domains

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Table 4. Characterization of Compounds 5−7

	Compound 5	Compound 6	Compound 7
FXI Activation Inhibition IC_{50} (nM); Ligand Efficiency (LE)	54; 0.24	42; 0.23	20; 0.25
Human 1.5x aPTT / Human PT (μM)	14/ >100	1.4/ >100	0.5/ >100

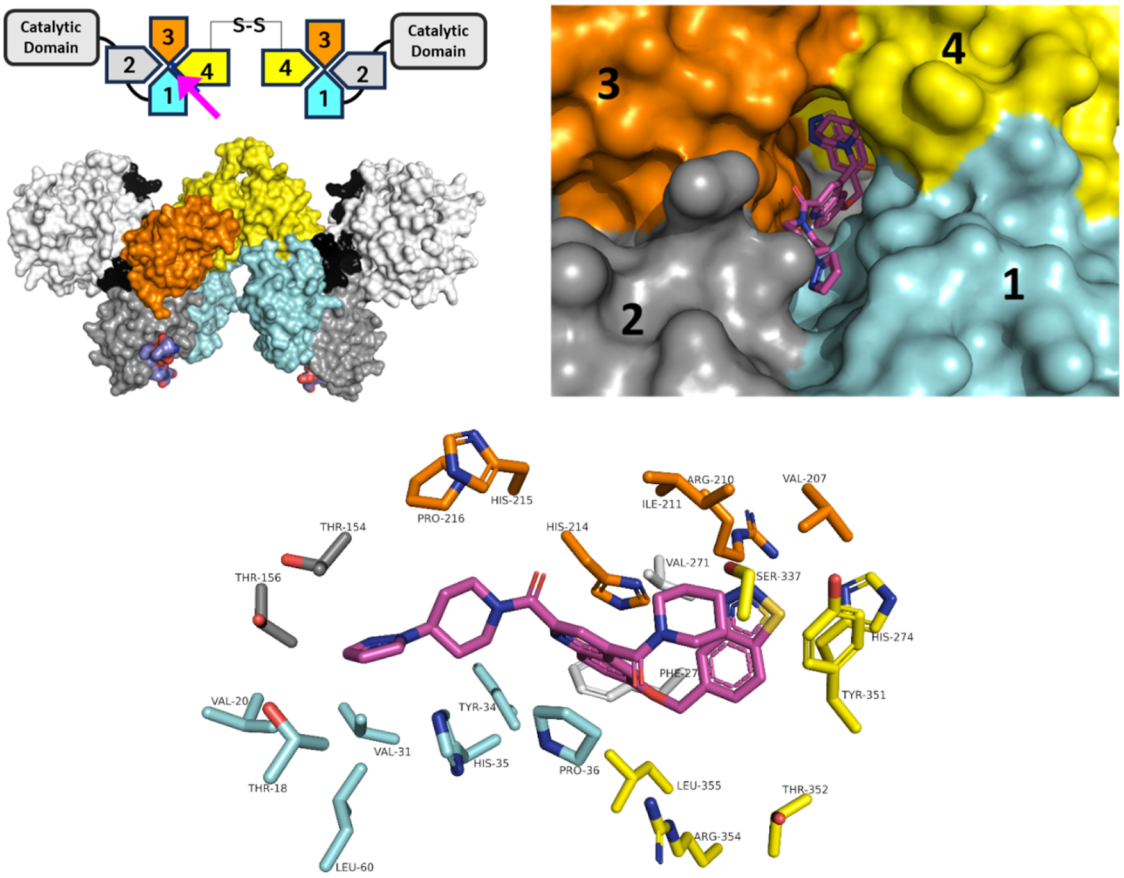


Figure 5. X-ray structure of FXI-bound compound 7. A cartoon graphic illustrating the binding site at the junction of the Apple domains (see the pink arrow) is shown in the upper left, along with the residues proximal to compound 7 within the binding pocket.[45]

(see the Pink Arrow in Figure 5). It should be noted that despite providing structural confirmation that XIAIs exhibit a specific mechanism for inhibiting FXI activation, a precise connection of XIAI binding to a conformational effect to preclude FXI activation remains unknown.

Building on the in vitro activity in the aPTT coagulation assay, evaluation of compound 7 in a translatable in vivo model remained as a key validation step for this mechanism. In vivo antithrombotic activity employed a rabbit acute pharmacodynamic model involving vascular placement of an extracorporeal arteriovenous (AV) shunt between the femoral artery and vein, respectively, which is well precedented and has been correlated with efficacy in humans.[47] Specifically, in the rabbit model of thrombosis, quantitative inhibition data is derived from weight measurement of thrombus deposition on a multifilament braided silk thread (serving as a nidus for thrombus formation) that is contained within the AV shunt construct. Evaluation of compound 7 as an IV infusion following an initial IV bolus loading dose led to the observation that an efficacious dose could be achieved at 2 mpk, wherein roughly 50% clot weight reduction was observed (Figure 6). This confirmed the viability of small-molecule FXI Activation Inhibitors as a novel mechanism for modulating hemodynamic effects.

DISCUSSION

Despite recent regulatory approval of direct thrombin and Factor Xa inhibitors,[9−11] the search for new and safer antithrombotics remains as a challenge in drug discovery.[12,13,16] Significant effort has been made to interrogate the FXI/FXIa node within the intrinsic coagulation cascade, leading to the discovery of a number of reported small molecule and biologic agents.[22−38] We report herein the

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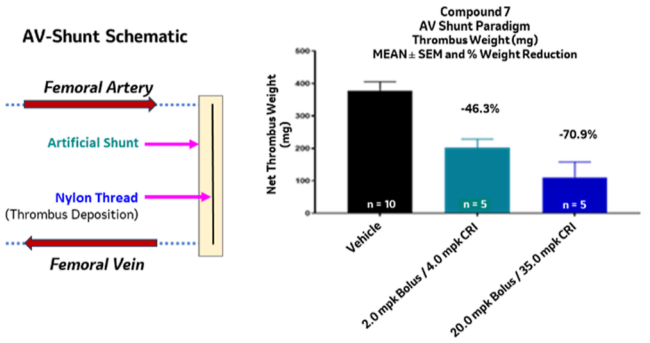


Figure 6. Schematic for the AV shunt used in the rabbit pharmacodynamic model (left), and the efficacy dose response of compound **7** in the rabbit AV shunt model of thrombosis (right).[48]

discovery of a novel mechanism for inhibiting the action of FXI, which combines the properties of these two modalities, specifically, that we have identified small molecules that inhibit FXI/FXIa in a manner similar to that previously reserved for antibodies by preventing FXI activation by an upstream effector, FXIIa.

Identification of potential candidate small molecules that could affect such pharmacology was initiated from an AS-MS affinity-based screen that identified hits that bound to the full-length FXI zymogen. At this point, we had not confirmed either the function or the binding site for any confirmed hit; however, we were able to develop a linked enzymatic assay that could establish the former, which was a necessary first step in confirming our proposed pharmacology. The lack of activity in functional counterscreens against both activated FXIa, as well as the effector FXIIa, provided additional evidence that inhibition of FXI Activation proceeded through the unmodified zymogen.

Direct binding experiments via SPR indicated saturable binding of the zymogen, which affirmed specific inhibition by the XIAIs. In addition, the SPR binding data suggested that the hit compounds subdivided into two distinct pharmacologies wherein compounds **1** and **2** exhibited faster k_{on}/k_{off} binding kinetics compared with compounds **3** and **4**, despite little difference in K_d, nor IC_{50} potency in the FXI Activation Assay. Gratifyingly, the relevance of this distinction became evident when each hit was evaluated for the functional inhibition of coagulation in plasma measured by the aPTT assay. The fact that compounds **3** and **4** inhibited coagulation may be derived from their longer FXI residence times that can be inferred from their off-rates and conceptually aligns with the high enzyme occupancy required for inhibiting the activated proteases within either coagulation cascade. Additionally, **3** and **4** are statistically distinct from **1** and **2**, which exhibit similar levels of plasma protein binding[49] (Table 3). Taken to the extreme, coagulation cascade amplification is possibly affected by a single protease molecule that escapes inhibition, and while practically, this simply requires enzyme occupancy >99%, the longer FXI zymogen residence time functions similarly to prevent coagulation cascade amplification, and hence fibrin clot formation. This distinction enabled prioritization of the XIAI hits for series expansion and a forward medicinal chemistry focus.

Early Hit-to-Lead efforts enabled facile SAR evolution around the common quinoline core scaffold from diamides **3** and **4** to afford XIAIs **5** and **6**, which exhibited good potency and ligand efficiency that were comparable to those of our initial hit compounds that perhaps revealed limitations of the linked enzymatic assay to finely differentiate XIAIs. More informative to the overall tractability was the dramatic improvement in aPTT potency while simplifying the core substituents at C4 and C6, which strongly supported the tractability for the quinoline diamide series and provided a more relevant activity correlate. One explanation for this disconnection was that the linked enzymatic assay, while nicely able to confirm the desired on-target function for the XIAIs, still represented a contrived system that could not replicate the complicated coagulation dynamics of the aPTT and PT assays.

Further SAR refinement led to the discovery of compound **7**, which could be leveraged to bridge two key gaps in validating this novel mechanism, specifically, that diamide **7** was cocrystallized with the full-length FXI zymogen to illustrate the binding site. Examination of the cocrystal structure of **7** with the zymogen shows little main chain movement of the Apple domains compared to the previously published apo structure.[19] However, compound **7** makes contacts with all four Apple domains, which is unique to this class of allosteric inhibitors, starting with the interaction of the pyrazolyl piperazine moiety contacting Apple domains 1 and 2, while the remainder of the compound chain extends into a space that enables contact with Apple domains 3 and 4 (Figure S5). Despite a lack of main chain movement, it should be noted that the observed binding pocket is absent from the apo zymogen structure, indicating an induced fit. Further, the binding pocket is distal to the active site, such that one can envision an induced fit binding event to the activated enzyme, whereby function is derived from preactivation zymogen binding. A deeper investigation would be required to inform on a detailed mechanism of action. Regardless, the bound structure of **7** provides direct evidence of a discrete bimolecular complex that results in inhibition of FXI activation.

Given the challenges with translating any related FXI expression levels or activity data sets to clinical bleeding outcomes, we opted to forego deeper investigation of bleeding at this early stage. Human FXI deficiency rarely results in spontaneous bleeding, although deficient individuals do present tissue-specific bleeding following surgery or injury, occurring primarily at sites with high fibrinolytic activity (mouth, nose, genitourinary tract).[50−53] Neither FXI antigen nor activity correlates with clinical bleeding risk, and aPTT assays do not predict bleeding risk.

Lastly, the successful suppression of clot formation in the rabbit AV shunt assay translated the in vitro plasma functional activity to an in vivo efficacy. Although a potential role of thrombin activation of FXI via a well-known feedback mechanism may occur,[17,18] this required detailed biophysical investigation beyond the scope of this project and was further ameliorated by the absence of activity by compounds counterscreened against a panel of coagulation factors (Table S1). Demonstration of PD efficacy in a commonly accepted model provided additional validation for XIAIs as an antithrombotic mechanism.

The discovery of the XIAIs represents a novel mechanism for drugging the FXI/FXIa locus and the intrinsic coagulation cascade, which continues to be a potential avenue for extending the utility of antithrombosis treatments. Following an affinity-based primary screen with a specifically designed functional assay amenable to high throughput provided an efficient hit triage process that enabled the identification of

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multiple structurally distinct XIAIs that possess unique pharmacology. Further investigation into the precise mechanism of action as well as progression toward compounds suitable for development will be the subject of future publications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsptsci.5c00265.

Supplementary tables and figures as well as synthetic methods and characterization for compounds **1**−**7** (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

AS-MS, affinity selection-mass spectrometry; DMSO, dimethyl sulfoxide; FXIIa, Factor XIIa; IV, intravenous injection; LCMS, liquid chromatography−mass spectrometry; mpk, milligrams per kilogram; NMR, nuclear magnetic resonance; PD, pharmacodynamic (efficacy); SDS-PAGE, sodium dodecyl sulfate−polyacrylamide gel electrophoresis; SPR, surface plasmon resonance; TFA, trifluoroacetic acid; TFF, tangential flow filtration; TLC, thin layer chromatography; XIAI, Factor XI Activation Inhibitor

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